

# Final Report

This project explores the use of machine learning, specifically the Light Gradient Boosting Machine to predict salaries based on various categorical and numerical factors. The goal is to assess the predictive performance of LightGBM and examine how its hyperparameters affect generalization through simulation.

The dataset from aijobs.net contains salary information for tech workers, with features such as job title, experience level, employment type, company size, remote work ratio, employee residence, and company location. All of these features were categorical, and my goal was to do regression so I preprocessed the data using target encoding. This method replaces categorical values with the mean salary for each category. The data was split into 70% training 15% validation, and 15% testing sets.

The model that I ended up working with was LightGBM due to its efficiency and strong performance on structured datasets. It builds trees in a leaf-wise manner and handles categorical features well after encoding. The model was trained using a regression objective to predict salary as a continuous variable. The model achieved an training RMSE of 3,730.67, validation RMSE of 9,246.20, and a testing RMSE of 8,274.13. This means that on average, the salary predictions were approximately \$8,274.14 off from the actual values during testing. These values indicate good performance, though some overfitting is evident from the training-validation gap. Feature importance revealed that job title and employee residence were among the most influential features, while employment type had minimal impact.

To evaluate the robustness of the model, I performed a Monte Carlo simulation to choose the best hyperparameters. Multiple synthetic datasets were generated to resemble the original dataset, and LightGBM models were trained across different hyperparameter combinations. For each combination, RMSE scores were recorded across 10 simulations. The best performing combination of hyperparameters had an RMSE of 10,160.38.

In conclusion, the project partially answered the research question. The LightGBM model achieved a testing RMSE of \$8,274.13, suggesting it performs reasonably well in predicting salaries based on features like job title, experience level, and company size. This level of error is relatively good, considering the complexity and variability of salary data. However, the variability across simulation results indicates that the model's generalizability could still be improved. Additionally, while target encoding made the categorical features compatible with

LightGBM, it may not be the most robust method as it can introduce data leakage or overfitting if not carefully applied. Ethically, any salary prediction model must be used cautiously to avoid perpetuating bias or unfair wage assumptions, especially in sensitive decision-making contexts.